TSPol-ASLIC: Adaptive Superpixel Generation With Local Iterative Clustering for Time-Series Quad- and Dual-Polarization SAR Data

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***Abstract*— The superpixel generation is a key step for object-based classification and change detection. For the time-series polarimetric synthetic aperture radar (PolSAR) superpixel generation, the traditional polarimetric similarity measure based on the joint covariance matrix has limitations in discriminating different time-series similarity sequences with different fluctuations. Besides, in the traditional time-series PolSAR superpixel generation methods, it is difficult to determine the tradeoff factor between polarimetric and spatial similarity. In this article, an adaptive time-series PolSAR superpixel generation method based on the simple local iterative clustering (SLIC) is proposed, named time-series polarimetric SAR (TSPol)-adaptive simple local iterative clustering (ASLIC). There are three main improvements. First, a novel time-series polarimetric similarity measure based on the root mean square (rms) is proposed. Multitemporal polarimetric statistical information is combined to describe the polarimetric proximity between pixels, referring to the rms of the multitemporal proximities. Second, an edge detection method based on the stacked 2-D Gaussian-shaped (s2-D GS) window is proposed to initialize the central seeds for superpixel generation. Third, an improved SLIC clustering similarity combined with the time-series polarimetric, time-series power, and spatial similarities is proposed. Meanwhile, a homogeneity factor is applied to adaptively balance the relative weights of various similarities. We use eight Radarsat-2 quad-polarization synthetic aperture radar (SAR) images and 14 Sentinel-1 dual-polarization SAR images to evaluate the effectiveness. The results show our similarity measure and superpixel generation results are superior to those of the traditional methods. For example, as for the Radarsat-2 data, the improvement of the boundary recall by the proposed similarity measure and homogeneity factor is about 4% and 10%, respectively.**

***Index Terms*— Polarimetric synthetic aperture radar (PolSAR), simple local iterative clustering (SLIC), superpixel generation, time series, time-series polarimetric SAR (TSPol)-adaptive simple local iterative clustering (ASLIC).**

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I. Introduction

SUPERPIXEL generation is one of the most important technologies for polarimetric synthetic aperture radar (PolSAR) image processing. It can not only reduce the effect of speckle noise but also aggregate similar neighboring pixels into homogeneous regions, making full use of their spatial and statistical information. Compared with traditional pixel-based methods, superpixel-based methods improve the accuracy of remote sensing thematic maps, and increase the efficiency of image processing [1], [2]. Such methods have been widely applied in image classification [3], [4], target detection [5], [6], and time-series monitoring [7]–[9].

With the enrichment of satellite platforms and the shortening of the revisit cycle, more and more time-series PolSAR datasets are available [10], such as Sentinel-1A/B (12 days for the single satellite and up to six days for the combination of two satellites) [11], TerraSAR-X/TanDEM-X (11 days for the repeat-pass mode of single satellite and 10 s for the pursuit monostatic mode of two satellites) [12], COSMO-SkyMed (16 days for single satellite and up to one day for the entire constellation) [13] and RADARSAT Constellation Mission (four days exact revisit for the coherent change detection with an interferometric synthetic aperture radar (InSAR) mode) [14]. Although abundant multitemporal PolSAR images promote the development of time-series classification and monitoring, they also increase the processing burden. In practice, many application scenarios require timely and effective data processing, such as the emergency response when geological disasters occur [15], and the determination of the yield loss by agricultural insurance institutions on large-scale disaster-stricken farmlands [16]. Therefore, it is necessary to develop an effective time-series PolSAR superpixel generation method, to assist in obtaining timely and accurate results.

The research studies of time-series PolSAR superpixel generation methods can refer to the object segmentation algorithms using single-date PolSAR images [17]. At present, the object segmentation methods have been developed a lot, including threshold-based segmentation [18], [19], graph theory-based segmentation [20]–[22], and region-based segmentation methods [23]–[25]. It should be noted that the region-based segmentation methods have attracted a lot of attention as it is easy to understand and efficient. Such methods use the region growing algorithm to iteratively expand homogeneous regions. And the representative algo-

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rithms include the simple local iterative clustering (SLIC) [26], the watershed [27], and the meanshift [28] algorithms. Among them, the SLIC algorithm has achieved rapid developments in PolSAR superpixel generation with less manual intervention and easy accessibility [1], [29], [30]. The PolSAR SLIC algorithms improve the similarity distance in k-means clustering with polarimetric information. In detail, such common polarimetric information include: 1) the RGB color information of the polarimetric Pauli color-coded], [32]; 2) the Wishart distance and its improved version derived from the polarimetric statistical hypothesis test theory [25], [26], [33], [34]; and 3) polarimetric decomposition scattering power or other features [35]. According to robust polarimetric statistical distribution, the similarity based on the Wishart distribution has been most widely used. As for the time-series PolSAR data, multitemporal covariance matrices can be merged into a joint covariance matrix [36]. Alonso-González *et al.* [37] described the applications of the joint covariance matrix in the binary tree partition tree (BPT), which is defined as the temporal evolution BPT. The segmentation results based on the BPT have proved that the joint covariance matrix has the potential to characteristics temporal evolutions and the time-series polarimetric similarity is essential for the correct clustering of pixels. Bao *et al.* [38] directly applied such joint covariance matrix obeyed by the Wishart distribution to generate the time-series polarimetric similarity. It is then used in the SLIC segmentation framework to generate superpixels with two quad-polarization images. However, such similarity used by Bao essentially is a linear sum of multiple single-date similarities, that has limitations in separating different time-series similarity sequences with different fluctuations. It constrains the applicability of the polarimetric similarity in accurately characterizing temporal evolutions of targets. This point will be clarified in Section II. Besides, in the existing time-series PolSAR superpixel generation methods, it is hard to determine the tradeoff factor between polarimetric and spatial similarities in the SLIC clustering.

Therefore, an adaptive time-series PolSAR superpixel generation method is proposed in this article, named time-series polarimetric SAR (TSPol)-adaptive simple local iterative clustering (ASLIC). The main contributions are listed as follows.

1) First, an improved time-series polarimetric similarity measure based on the root mean square (rms) is proposed. Such similarity expands the symmetric revised Wishart distance and uses the rms to describe the temporal evolutions of time-series polarimetric similarities.
2) Second, an edge detection method based on the stacked 2-D Gaussian-shaped (s2-D GS) window is proposed. Both the spatial and temporal neighborhood information are considered simultaneously, to generate the complete and accurate boundaries, which are beneficial to the initialization of the central seeds for the superpixel generation.
3) Third, the time-series polarimetric, time-series power, and spatial similarities are combined to describe the SLIC clustering proximity between superpixels. For adaptively controlling the relative weights between the polarimetric and spatial similarity, the time-series homogeneity factor is defined based on the local homogeneity degree and the edge strength.

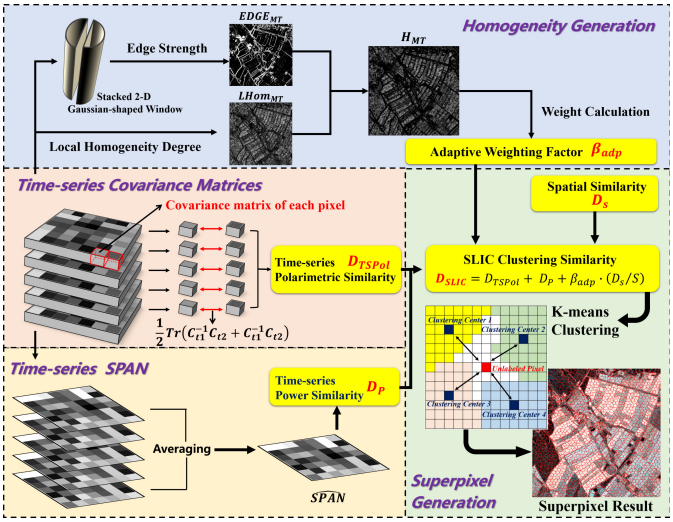

Fig. 1. Implementation procedure of the TSPol-ASLIC method.

This article is organized as follows, the TSPol-ASLIC superpixel generation method is described specifically in Section II. The study area and the utilized time-series PolSAR datasets are introduced in Section III. In Section IV, the edge detection and superpixel generation results are analyzed and discussed by visual presentation and quantitative evaluation. Specifically, we compare the TSPol-ASLIC method and the Pol-ASLIC method, discuss various combinations of time-series polarimetric similarities and the SLIC clustering similarities, then analyze the improved effects of the homogeneity factor and the time-series power similarity. The conclusions are revealed in Section V.

II. TSPol-ASLIC Superpixel Generation

Referring to the SLIC superpixel generation framework [39] (a. clustering center initialization, b. k-means clustering in a local area, and c. postprocessing), the implementation procedure of the proposed TSPol-ASLIC method is shown in Fig. 1. It can be seen that the key points of our method are the time-series polarimetric similarity, the homogeneity factor, and the SLIC clustering similarity, which will be described in detail as follows.

A. Similarity Measure Between the Two Sets of Time-Series Polarimetric Covariance Matrices

The similarity measure between pixels in time-series PolSAR images is the essential basis of our proposed segmentation framework. An effective similarity measurement can not only describe the polarimetric distance between pixels but also reflect the temporal evolution of the target. This section first describes the revised Wishart distance, then introduces two kinds of distance measure between time-series covariance matrices. One is the conventional similarity measure based on the joint covariance matrix, the other is the proposed similarity measure based on the rms.

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As for a reciprocal target, the multipolarization information can be written with a complex scattering vector k based on the Lexicographic basis as

$$k = \begin{cases} [S_{\mathrm{HH}}, S_{\mathrm{HV}}, S_{\mathrm{VV}}]^T, & \mathrm{Quad-pol(full)} \\ [S_{\mathrm{HH}}, S_{\mathrm{HV}}]^T, & \mathrm{Dual-pol(pp1)} \\ [S_{\mathrm{VH}}, S_{\mathrm{VV}}]^T, & \mathrm{Dual-pol(pp2)} \\ [S_{\mathrm{HH}}, S_{\mathrm{VV}}]^T, & \mathrm{Dual-pol(pp3)} \end{cases} \tag{1}$$

where S_{HH}, S_{HV}, and S_{VV} are the elements of the complex backscattering matrix, H and V denotes the horizontal and the vertical channel, respectively, T means the transposition operator. Particularly, there are three modes for the dual-polarization data, including pp1 (HH + HV), pp2(VH + VV), and pp3(HH+VV), pp means the dual-polarization mode [40].

For the sake of describing the statistical characteristics of the distributed targets, the complex scattering vector is always converted into its second-order statistic, such as the polarimetric covariance or coherency matrix. Considering the Lexicographic basis [41], such matrix can be described by the polarimetric covariance matrix C as

$$C = \langle kk^{\dagger} \rangle \tag{2}$$

where $\langle \cdot \rangle$ denotes the samples averaging, $\dagger$ denotes the conjugation and transposition operator. In terms of the PolSAR image, the Wishart function is used to describe the statistical distribution of matrix C [42]. Referring to [43], the corresponding likelihood ratio function is formulated to test the similarity between two regions. The hypothesis test is

$$\begin{cases} H_0 : \Sigma_i = \Sigma_j \\ H_1 : \Sigma_i \neq \Sigma_j \end{cases} \tag{3}$$

where Σ_i and Σ_j are the distribution parameters in the Wishart function. They are always replaced by their maximum likelihood estimation, which are the center covariance matrices of the ith and the jth region [43]. Hence, the similarity can be calculated by the generalized likelihood ratio test Q_1 as [26], [44], [45]

$$\begin{aligned} Q_1 &= \frac{L_{H_0}\left(\hat{\Sigma}_j \mid \Theta_i\right)}{L_{H_1}\left(\hat{\Sigma}_i \mid \Theta_i\right)} = \frac{\prod_{n=1}^{N_i} p\left(C_n \mid L, \hat{\Sigma}_j\right)}{\prod_{n=1}^{N_i} p\left(C_n \mid L, \hat{\Sigma}_i\right)} \\ &= \frac{\left|\hat{\Sigma}_i\right|^{LN_i}}{\left|\hat{\Sigma}_j\right|^{LN_i}} \exp\left\{-LN_i \mathrm{Tr}\left(\hat{\Sigma}_j^{-1}\hat{\Sigma}_i\right) - m\right\} \end{aligned} \tag{4}$$

where L denotes the number of looks, N_i and N_j mean the number of samples within two regions Θ_i and Θ_j, $\hat{\Sigma}_i$ and $\hat{\Sigma}_j$ are the average covariance matrices of two regions, $\mathrm{Tr}(\cdot)$ means the trace of a matrix. m is the dimension of k, m equals to three for the quad-polarization data, and m equals two for the dual-polarization data. After maximizing the likelihood ratio Q_1, the symmetric revised Wishart distance can be defined [44] as

$$D_{\mathrm{SRW}}\left(\Theta_i, \Theta_j\right) = \frac{1}{2}\mathrm{Tr}\left(\hat{\Sigma}_j^{-1}\hat{\Sigma}_i + \hat{\Sigma}_i^{-1}\hat{\Sigma}_j\right) - m. \tag{5}$$

D_{SRW} is a robust distance measure between covariance matrices. It satisfies all general metric conditions, except the triangle inequality [44]. Furthermore, it has been widely applied to the segmentation and classification fields [26], [44]. Based on the D_{SRW}, the two kinds of time-series polarimetric similarities are introduced as follows.

1) Conventional Time-Series Polarimetric Similarity With the Joint Covariance Matrix: Under the assumption that the multitemporal data are uncorrelated, time-series polarimetric covariance matrices can be combined into a joint covariance matrix [36], [46] as

$$Z = \begin{bmatrix} \hat{\Sigma}_{t1} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \hat{\Sigma}_{tN} \end{bmatrix}_{mN \times mN} \tag{6}$$

where $\hat{\Sigma}_{ti}$ denotes the ith-date covariance matrix, N means the number of dates, and $\mathbf{0}$ is a $m \times m$ matrix where all elements are zero. It should be noted that such assumption of uncorrelation is only an approximation for most targets, unless the temporal baseline is very large. Such joint covariance matrix also obeys the Wishart distribution, so that the symmetric revised Wishart distance between two joint covariance matrices can be written as

$$D_Z\left(\Theta_i, \Theta_j\right) = \frac{1}{2}\mathrm{Tr}\left(\hat{Z}_j^{-1}\hat{Z}_i + \hat{Z}_i^{-1}\hat{Z}_j\right) - mN \tag{7}$$

where $\hat{Z}_i$ and $\hat{Z}_j$ is the average joint covariance matrix of two regions Θ_i and Θ_j, respectively. Because such joint covariance matrices are the block-diagonal matrices, the D_Z can be written as the sum of multiple single-date D_{SRW} (see Appendix A) as

$$D_Z = \sum_{t=1}^{N} D_{\mathrm{SRW}}^t = \sum_{t=1}^{N}\left[\frac{1}{2}\mathrm{Tr}\left(\hat{\Sigma}_{tj}^{-1}\hat{\Sigma}_{ti} + \hat{\Sigma}_{ti}^{-1}\hat{\Sigma}_{tj}\right) - m\right] \tag{8}$$

where D_{SRW}^t denotes the revised Wishart distance of the tth date. Then the temporal average operation is carried out to weaken the influence of the number of dates on the range of polarimetric similarities. So that, the conventional time-series polarimetric similarity can be written as

$$D_{\mathrm{TSPol}}^{J\mathrm{mat}}\left(\Theta_i, \Theta_j\right) = \frac{1}{N}\sum_{t=1}^{N}\frac{1}{2}\mathrm{Tr}\left(\hat{\Sigma}_{tj}^{-1}\hat{\Sigma}_{ti} + \hat{\Sigma}_{ti}^{-1}\hat{\Sigma}_{tj}\right) - m. \tag{9}$$

However, such traditional similarity is essentially the direct sum of multitemporal similarities, which has constraints in separating different time-series similarity sequences with different fluctuations. For example, the sequence with large fluctuations and the sequence with small fluctuations may have the same average value. This limits the development of time-series polarimetric similarity in distinguishing the temporal evolutions. Based on this, a novel similarity measure is proposed.

2) Novel Time-Series Polarimetric Similarity Based on the RMS: Different from the linear stack and averaging process of the aforementioned conventional similarity, the novel similarity characterizes the multitemporal similarities based on the rms. Compared with the mean value of the multitemporal similarity sequence, the rms of such sequence can enlarge the weight of the large similarity distances in the sequence and highlight the date with the obvious change (as shown

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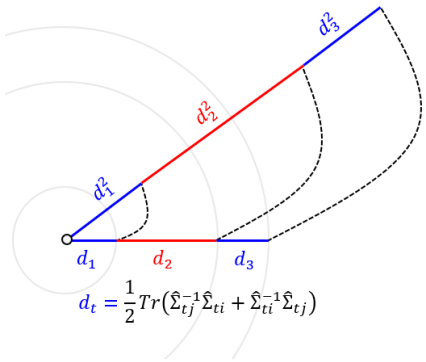
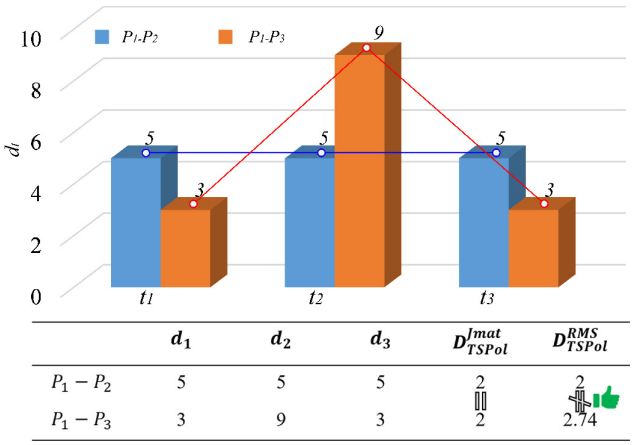


Fig. 2. Difference of the two kinds of time-series polarimetric similarity $D_{\mathrm{TSPol}}^{J\,\mathrm{mat}}$ and $D_{\mathrm{TSPol}}^{\mathrm{rms}}$ under three dates. It can be found that the square sum of d_1, d_2, and d_3 enlarges the weight of the large similarity (d_2), and highlights the date with obvious variation.



	d_1	d_2	d_3	D_{TSPol}^{Jmat}	D_{TSPol}^{RMS}
$P_1 - P_2$	5	5	5	2	2
$P_1 - P_3$	3	9	3	2	2.74

Fig. 3. One case of comparison between the two kinds of time-series polarimetric similarities.

in Fig. 2). It is more sensitive to the fluctuations of the similarity sequence than the mean value, which is beneficial to separate the sequence with small and gentle fluctuations and the sequence with large and sharp fluctuations. It makes the rms more suitable for the description of the temporal evolutions of similarities than the mean value, which is beneficial for the correct clustering of pixels in time-series PolSAR segmentation. With multitemporal covariance matrices, the novel similarity can be written as

$$D_{\mathrm{TSPol}}^{\mathrm{rms}}(\Theta_i, \Theta_j) = \sqrt{\frac{1}{N}\sum_{t=1}^{N}\left(\frac{1}{2}\mathrm{Tr}\left(\hat{\Sigma}_{tj}^{-1}\hat{\Sigma}_{ti} + \hat{\Sigma}_{ti}^{-1}\hat{\Sigma}_{tj}\right)\right)^2} - m. \tag{10}$$

Furthermore, three pixels (P_1, P_2, and P_3) with different time-series polarimetric covariance matrices are defined to explain the advantage of the $D_{\mathrm{TSPol}}^{\mathrm{rms}}$. As shown in Fig. 26 (see Appendix B), the corresponding covariance matrices are denoted with the polarization signatures [47], [48], which visually presents the polarimetric characteristics with the normalized copolarization and crosspolarization powers under various polarization bases [48]. Specifically, the polarization signatures of P_1 and P_2 in three dates are close to the dipole scattering targets, and their temporal variations are similar. Different from $P_1 - P_2$, the covariance matrices of P_1 and P_3 in the first and third dates are the same, but their matrices on the second date are obviously different. The polarization signatures of P_3 in the second date is close to the surface scattering target. The time-series similarities d_1, d_2, and d_3 of $P_1 - P_2$ are 5, 5, and 5, respectively, where $d_t = (1/2)\mathrm{Tr}(\hat{\Sigma}_{tj}^{-1}\hat{\Sigma}_{ti} + \hat{\Sigma}_{ti}^{-1}\hat{\Sigma}_{tj})$. The corresponding similarities of $P_1 - P_3$ are 3, 9, and 3, respectively. The $P_1 - P_2$ is used to simulate the similarity sequence with small fluctuations, and the $P_1 - P_3$ corresponds to the sequence with large fluctuations. As shown in Fig. 3, the $D_{\mathrm{TSPol}}^{\mathrm{rms}}$ can effectively distinguish two pairs of pixels, while the conventional similarity $D_{\mathrm{TSPol}}^{J\,\mathrm{mat}}$ determines that two pairs of pixels are the same. It proves that our proposed similarity can improve the separated degree of similarity sequences with different time-series variations. It has the potential to distinguish various temporal evolutions of targets.

B. Homogeneity Factor Based on the Time-Series PolSAR Edge Extraction Method

The homogeneity factor suitable for the time-series PolSAR data is formulated in this section. Referring to [26], such factor is an enhanced homogeneity measure, which is the fraction between the local homogeneity degree (LHom) and the edge strength (EDGE), where the LHom is indicated with the equivalent number of look (ENL) in the single-date PolSAR data. ENL is the key indicator in polarimetric statistical theory [41]. It can be used to indicate the speckle noise level and evaluate the homogeneity degree [49], which describes the spatial variation of the local textural region. Referring to [49], we sum the traces of multitemporal covariance matrices and calculate the LHom of the time-series PolSAR datasets based on the trace moment estimation theory [50] as

$$\mathrm{LHom}_{\mathrm{MT}} = \frac{\sum_{t=1}^{N}\mathrm{Tr}(\Sigma)^2}{\sum_{t=1}^{N}\langle\mathrm{Tr}(C_p C_p)\rangle - \sum_{t=1}^{N}\mathrm{Tr}(\Sigma\,\Sigma)} \tag{11}$$

where Σ is the average covariance matrix of one region, C_p is the covariance matrix of a single pixel. $\mathrm{LHom}_{\mathrm{MT}}$ considers the multitemporal polarimetric information simultaneously, and achieves more robust statistical characteristics than those of single-date PolSAR image [49]. It can distinguish different textural regions and extract the parcel boundaries.

An accurate edge strength map provides pixel growth constraints during the superpixel generation. Referring to the edge detection method in [26], the polarimetric gradients along different directions in a GS window should be calculated first. To explore the time-series PolSAR data, we modify the 2-D GS window into the stacked 2-D (s2-D) GS window, which is the stack of multiple 2-D GS windows in the temporal dimension. As shown in Fig. 4, one s2-D GS window includes two semicircle subwindows [51], and can be described by five parameters, including the length R_1, the width R_2, the spacing d_f between two subwindows, the angular increment θ_f indicating the rotation of GS window [26], [51] and the number of time-series images N. Based on the s2-D GS window, two average covariance matrixes within two subwindows on both sides of the center pixel are estimated [26]. Furthermore, the polarimetric similarity D_{SRW} between such two average

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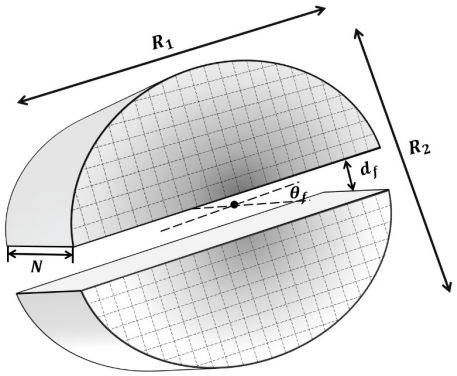

Fig. 4. Representation of the s2-D GS window.

covariance matrixes is calculated as the polarimetric gradient of this center pixel. Finally, the edge strength is confirmed as the maximum gradient among different directions of the filtering window [26]. It is worth noting that the s2-D GS window is valid for the targets with the stationary label during a period, such as the crops, of which types can remain unchanged from sowing to harvesting.

Specifically, the single 2-D GS window can be described as a Gaussian weighting matrix. For example, the weight of the pixel (x, y) in the horizontal 2-D window can be expressed as [52]

$$W(x, y) = \frac{1}{\sqrt{2\pi}\sigma_x\sqrt{2\pi}\sigma_y}\exp\left(-\left(\frac{x^2}{2\sigma_x^2}+\frac{y^2}{2\sigma_y^2}\right)\right) \quad (12)$$

where σ_x and σ_y controls the length R_1 and width R_2 of GS window, respectively. Based on the spacing d_f, such weighting matrix can be separated into two sub-matrixes, of which the borders are the cutoff Gaussian weights. Then the combination of multitemporal Gaussian weighting matrices forms the s2-D GS window. As for various directions, we rotate the s2-D GS window in the horizontal direction with various degrees to formulate a set of s2-D GS windows [51]. As for each s2-D GS window, the average polarimetric covariance matrix $\hat{\Sigma}_M$ can be obtained as

$$\hat{\Sigma}_M = \frac{\sum_{t=1}^{N}\sum_{(x,y)} W(x, y)C(x, y)}{N\sum_{(x,y)} W(x, y)}. \quad (13)$$

Then we calculate the D_{SRW} of two average polarimetric covariance matrices in two subregions within the s2-D GS window in each direction, and choose the maximum gradient as the edge strength. Finally, the edge detection result is generated by traversing all pixels. The time-series polarimetric edge detection method is also introduced in Algorithm 1. Based on the $\mathrm{LHom_{MT}}$ and the edge strength $\mathrm{EDGE_{MT}}$, the homogeneity factor H_{MT} can be calculated as [26]

$$H_{\mathrm{MT}} = \frac{\mathrm{LHom_{MT}}}{\mathrm{EDGE_{MT}}}. \quad (14)$$

Such fraction strengthens the homogeneity within the parcel and reduces the homogeneity of the boundary, helpful to adaptively choose the weights in different scenarios.

Algorithm 1 Time-Series PolSAR Edge Detection Method

Begin

1. Set the parameters σ_x, σ_y, θ_f and d_f.
2. For $N_f = \pi/\theta_f$ filtering window with different directions, calculate s2-D Gaussian weighting matrices, and calculate the average covariance matrices of two sub-regions using the formula *(13)*.
3. Calculate the D_{SRW} distance between two average covariance matrices on each direction according to the formula *(5)*.
4. Find the maximum distance D_{max} and the corresponding direction θ_{max}.
5. Save the distance D_{max} and θ_{max}, then move to the next pixel, and continue from the Step 2.

End

C. SLIC Clustering Similarity Measure of Time-Series PolSAR Data

Referring to [26], the SLIC clustering similarity for superpixel generation contains the time-series polarimetric similarity, the time-series power similarity, and the spatial similarity. Furthermore, the homogeneity factor is applied to balance the compactness and shape of the superpixels. The time-series polarimetric similarity is calculated by $D_{\mathrm{TSPol}}^{\mathrm{rms}}$ according to (10). Referring to the textural similarity in [26], the time-series power similarity can be described by the difference of the time-series average polarimetric total power between different superpixels as

$$D_P = \frac{\left|\hat{P}(x_i, y_i) - \hat{P}\left(x_j, y_j\right)\right|}{\max\left(\hat{P}\right)} \quad (15)$$

$$\hat{P} = \frac{1}{N}\sum_{t=1}^{N} P_t \quad (16)$$

where P_t denotes the total scattering power SPAN of the tth date. $\hat{P}$ denotes the average total power, and describes the total received signal of objects, which varies among different object types. For instance, the $\hat{P}$ of bare soil is lower than that of crop and that of the stable building is the highest. This kind of characteristic is helpful for the correct division of superpixels. For the sake of combination, the D_P should be normalized by the maximum of $\hat{P}$. Finally, the spatial similarity is defined as [39]

$$D_s = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}. \quad (17)$$

These three kinds of similarities are combined together to describe the all-round similarity for the SLIC clustering. Furthermore, the adaptive weighting factor is used for balancing relative weights of various similarities. Referring to the method in [26], the adaptive weighting factor β_{adp} is calculated with the H_{MT} as

$$\beta_{\mathrm{adp}} = \beta \cdot \frac{1}{2}\left(H_{\mathrm{MT}}(x_i, y_i) + H_{\mathrm{MT}}\left(x_j, y_j\right)\right) \quad (18)$$

where the parameter β can be set adaptively based on the local spatial complexity. Inspired by Yin *et al.* [45], we use the

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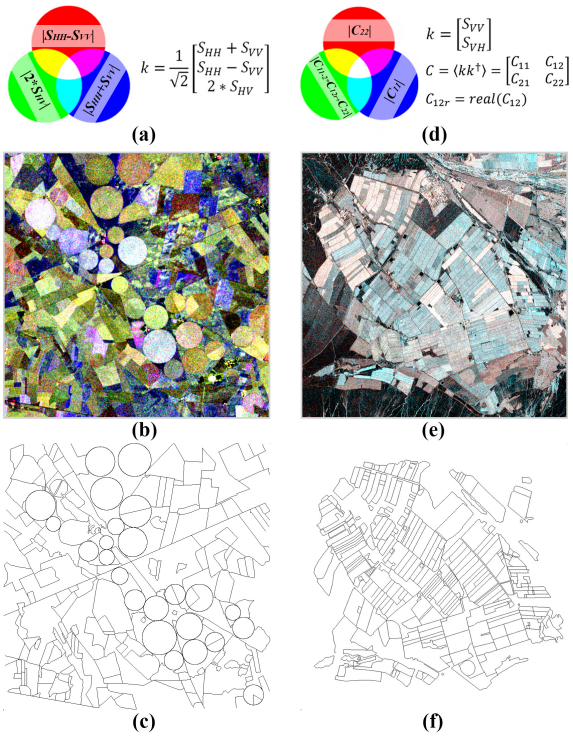

Fig. 5. Time-series average polarimetric color-coded image and the true edge map presentation. (a) Illustration of color-coded representation of the quad-polarization data. (b) Time-series average polarimetric color-coded image of the quad-polarization data. (c) True edge map corresponding to the quad-polarization data. (d) Illustration of color-coded representation of the dual-polarization data. (e) Time-series average polarimetric color-coded image of the dual-polarization data. (f) True edge map corresponding to the dual-polarization data.

form of direct sum of the time-series polarimetric, time-series power, and spatial similarities to define the SLIC clustering similarity as

$$D_{\mathrm{SLIC}} = D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_P + \beta_{\mathrm{adp}} \cdot (D_s / S) \tag{19}$$

where S is the sampling interval according to the number of superpixels. The β_{adp} can be adaptive to different scenes. In homogeneous regions, large β_{adp} increases the weight of spatial similarity to generate compact superpixels. On the contrary, the time-series polarimetric and time-series power similarity overweigh the spatial similarity in the complex heterogeneous region to make superpixel adhere to true boundaries.

III. Study Area and Datasets

This article selects eight quad-polarization Radarsat-2 synthetic aperture radar (SAR) images and 14 dual-polarization Sentinel-1 SAR images to evaluate the effectiveness of the proposed method. The time-series average polarimetric color-coded image and true edge map are shown in Fig. 5. It should be pointed out that the time-series Radarsat-2 data comes from the AgriSAR2009 campaign project supported by the European Space Agency (ESA) [53]. It covers the area in Barrax of southeastern Spain, which implements a uniform and centralized crop management, where the largest parcel has reached 1 km in diameter. And most of them are irrigated with circular irrigation system for saving water resources. Besides, the time-series Sentinel-1 data covers the area in Jinchang, China. The farmland here is also managed intensively, of which the average parcel size exceeds 100 m.

TABLE I
Satellite Imaging Parameters of the Quad- and Dual-Polarization Datasets

Satellite	Parameters	
Radarsat-2	Dates	20090403, 20090427, 20090521, 20090614, 20090708, 20090801, 20090825, 20090918.
	Polarimetric Mode	Quad-Polarization
	Center Incidence Angle	34.25°
	Band	C band
	Flight Direction	Ascending
	Range pixel spacing	4.7 m
	Azimuth pixel spacing	5.1 m
Sentinel-1	Dates	20190310, 20190322, 20190403, 20190415, 20190427, 20190521, 20190602, 20190614, 20190801, 20190813, 20190825, 20190906, 20190918, 20190930.
	Polarimetric Mode	VH+VV
	Center Incidence Angle	39.11°
	Band	C band
	Flight Direction	Ascending
	Range pixel spacing	2.33 m
	Azimuth pixel spacing	13.94 m

Throughout such two study areas, most of the crop types are single-season plants, with the main growth period ranging from March to September each year. Hence, we collect the available time-series PolSAR images as listed in Table I. All datasets should be preprocessed, including radiation calibration, time-series registration, and polarimetric filtering. It should be noted that the Sentinel-1 images have to be multilooked first, with 1×4 look numbers in azimuth and range direction, respectively. While the Radarsat-2 datasets have not been multilooked. Afterward, the refined Lee filtering is performed on all datasets with a window size of 5 [54].

IV. Results and Discussions

To demonstrate the effectiveness of our method, the edge detection, and superpixel generation results are discussed with visual presentation and quantitative evaluation. First, we present the edge detection results and compare superpixel generation results of our method using time-series PolSAR images and the results of the Pol-ASLIC method using single-date image [26]. Furthermore, different time-series polarimetric similarities and SLIC clustering similarities are discussed, the corresponding superpixel generation results are compared. For a clear comparison, Bao's method [38] is utilized as a representative of existing methods. It uses the conventional polarimetric similarity based on the joint covariance matrix and applies the sum of the polarimetric and spatial similarity to define the SLIC clustering similarity. Besides, the improvements of the proposed time-series homogeneity factor and the time-series power similarity on superpixel generation results are analyzed.

There are some parameters need to be clarified before discussions. For the edge detection algorithm (Algorithm 1),

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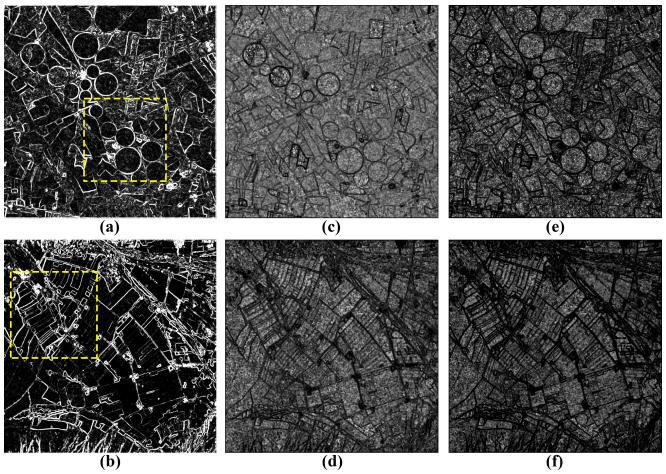

Fig. 6. Edge detection result, the $\mathrm{LHom_{MT}}$ result and the homogeneity result map generated by the proposed method with time-series quad- and dual-polarization data. (a) and (b) Edge detection results. (c) and (d) $\mathrm{LHom_{MT}}$ results. (e) and (f) Homogeneity result maps. The local edge detection results marked in yellow rectangle boxes in (a) and (b) will be clearly presented in Figs. 7 and 8, respectively.

the parameters σ_x, σ_y, θ_f, and d_f are set to 1.5, 1, $\pi/8$, and 1, respectively. The weighting parameter β in our method is set to 1, and the homogeneity factor is normalized by considering its distribution range. The tradeoff factor in Bao's method is set to 2.

To intuitively compare the difference between the TSPol-ASLIC and the Pol-ASLIC method, the visual presentations of the superpixel generation results are presented with the time-series average PolSAR data. The boundaries of superpixels are overplotted on the polarimetric color-coded image with the average data. There are two quantitative accuracy indicators being used, including the boundary recall (BR) and the achievable segmentation accuracy (ASA) [55]. Specifically, the BR is the overlap ratio between the superpixel boundary and the true ground boundary. The higher the BR, the better attachments between the superpixels and the real boundary. The ASA is the overlap ratio between superpixels and real segmentation objects. The higher the value of ASA is, the more superpixels fall into the correct segmentation objects. Therefore, the combination of such two indicators is suitable to judge the effectiveness of the superpixel generation method objectively.

A. Edge Extraction and Homogeneity Factor Generation Result

The edge detection results of our method and the single-date Pol-ASLIC method are compared, under the s2-D GS window and the 2-D GS window correspondingly. Fig. 6 presents the generated edge strength map, the local homogeneity degree ($\mathrm{LHom_{MT}}$) and the homogeneity result map (H_{MT}). The results show that the proposed edge detection method makes full use of the multitemporal information, and generates a clean edge detection result by pixel averaging under the s2-D GS window. As shown in Figs. 7 and 8, compared with the single-date edge detection method, the proposed edge detection method detects more real edges. As known to all, the complete and clear parcel boundaries will promote the superpixels attaching to the correct boundary. Furthermore, it illustrates that the homogeneity result map can well discriminate the parcel boundaries from the farmland area.

TABLE II

DIFFERENT COMBINATIONS OF TIME-SERIES POLARIMETRIC SIMILARITIES AND THE SLIC CLUSTERING SIMILARITIES AND THE DEFINITION OF VARIOUS COMBINATIONS

Method	Time-series Polarimetric Similarity	SLIC Clustering Similarity
Our Method	D_{TSPol}^{RMS} (with D_{SRW})	D_{SLIC}
$D_{TSPol}^{Jmat} + D_{SLIC}$	D_{TSPol}^{Jmat} (with D_{SRW})	D_{SLIC}
$D_{TSPol}^{Jmat} + D_{SLIC}^{2}$	D_{TSPol}^{Jmat} (with D_{SRW})	D_{SLIC}^{2}
$D_{TSPol}^{RMS} + D_{SLIC}^{2}$	D_{TSPol}^{RMS} (with D_{SRW})	D_{SLIC}^{2}
Bao's method [38]	$N \cdot D_{TSPol}^{Jmat}$ (with D_{RW})	D_{SLIC} ($\beta_{adp} = 2$ and without time-series power similarity)

Notes:

$$D_{RW} = \ln\left(\left|\widehat{\Sigma}_j\right|/\left|\widehat{\Sigma}_i\right|\right) + Tr\left(\widehat{\Sigma}_j^{-1}\widehat{\Sigma}_i\right) - m$$

$$D_{SLIC}^{2} = \sqrt{\left(D_{TSPol}{}^{2} + D_T^2\right) + \beta_{adp} \cdot (D_s/S)^2}$$

B. Comparison of Superpixel Generation Results Between the TSPol-ASLIC and the Pol-ASLIC Method

In this section, the superpixel generation results of the proposed TSPol-ASLIC method and the Pol-ASLIC method are compared, under different numbers of superpixels (K = 4000, 8000, 12 000, 16 000, 20 000). Three representative dates are selected for the Pol-ASLIC superpixel generation with single-date image, which represents the initial growth period, the rapid development period, and the mature period of most crops, respectively. Specifically, we select 20090427, 20090521, and 20090918 for Radarsat-2 images, and 20190322, 20190813, and 20190918 for Sentinel-1 images. As shown in Figs. 9–12, the superpixel generation results of our method are more actual than that of the Pol-ASLIC method, showing that the superpixel boundaries attach better to the true parcel boundaries, and the superpixels within each parcel are especially tight and regular. Fig. 13 presents the quantitative comparison. Our method achieves the highest BR and ASA accuracy under different numbers of superpixels. And when the number of superpixels is small, the improvement of our method is more obvious than the other cases.

C. Discussion and Analysis of the Time-Series Polarimetric Similarities and the SLIC Clustering Similarities

In this section, we compare and discuss the superpixel generation results with different time-series polarimetric similarities and SLIC clustering similarities. As shown in Table II, the two kinds of polarimetric similarities ($D_{\mathrm{TSPol}}^{\mathrm{rms}}$ and $D_{\mathrm{TSPol}}^{J\mathrm{mat}}$) and the two kinds of SLIC clustering similarities (D_{SLIC} and D_{SLIC}^{2}) are listed. Specifically, the D_{SLIC}^{2} uses the traditional combination method with the square sum of various similarities to define the SLIC clustering similarity, which is used

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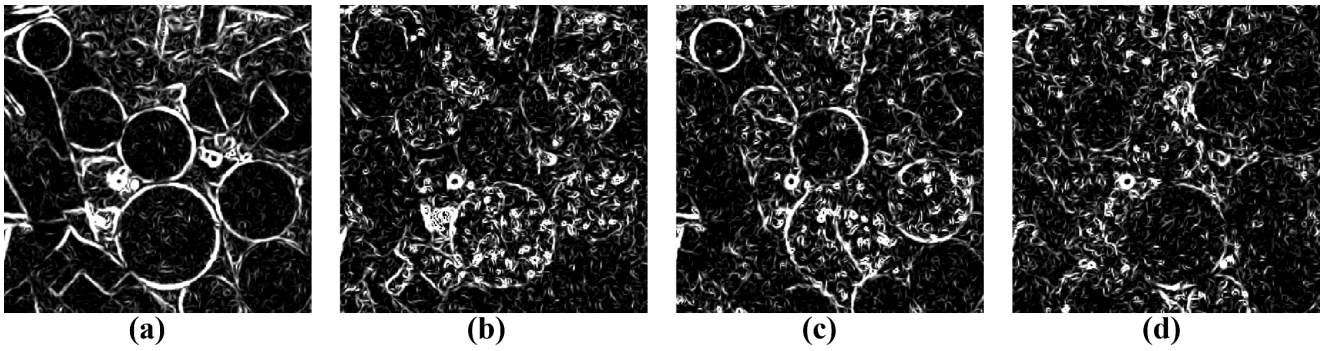

Fig. 7. Comparison of the local edge detection results [marked in the yellow rectangle box in Fig. 6(a)] between the proposed time-series detection method with the s2-D GS window and the single-date detection method with the 2-D GS window based on the Radarsat-2 images, (a) proposed method with time-series Radarsat-2 images, (b)–(d) denote the single-date edge detection results with the Radarsat-2 image in 20090427, 20090521, and 20090918, respectively.

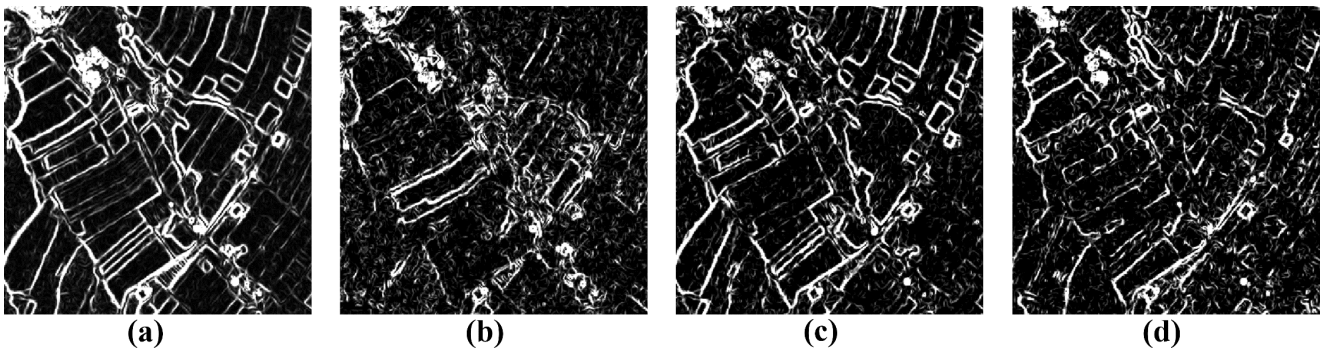

Fig. 8. Comparison of the local edge detection results [marked in the yellow rectangle box in Fig. 6(b)] between the proposed time-series detection method with the s2-D GS window and the single-date detection method with the 2-D GS window based on the Sentinel-1 images, (a) proposed method with time-series Sentinel-1 images, (b)–(d) single-date edge detection results with the Sentinel-1 image in 20190322, 20190813, and 20190918, respectively.

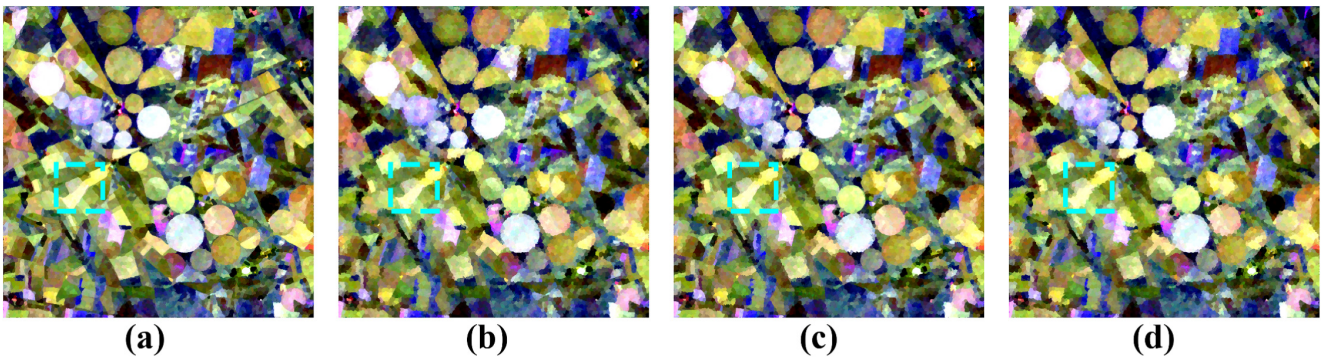

Fig. 9. Comparison of superpixel generation results ($K = 8000$) with the proposed method and the Pol-ASLIC method with the quad-polarization data. (a) Proposed method with the time-series images. (b) Pol-ASLIC method with the 20090427 image. (c) Pol-ASLIC method with the 20090521 image. (d) Pol-ASLIC method with the 20090918 image. The local results marked in cyan rectangle boxes in (a)–(d) will be clearly presented in Fig. 11. Particularly, the superpixel generation results are presented with the time-series average PolSAR data.

in many segmentation cases [25], [26], [39]. In addition, as a representative existing method, Bao's method is also compared [38]. It uses the conventional time-series polarimetric similarity with the joint covariance matrix. Particularly, different from Bao's method, our method uses the way of unitizing time-series polarimetric similarities. Such unitization changes the value of the similarity, but the relative relationship between similarities of different pixels is not changed. So that it does not affect the SLIC clustering process. As for the visual presentations, Figs. 14–21 present the comparisons of superpixel generation results generated by our method and other methods. The corresponding quantitative comparisons have been shown in Fig. 22. The detailed descriptions are presented as follows.

First, our method is compared with the method with $D_{\mathrm{TSPol}}^{J\mathrm{mat}} + D_{\mathrm{SLIC}}$, to clarify the advantage of the proposed time-series polarimetric similarity $D_{\mathrm{TSPol}}^{\mathrm{rms}}$. Our method generates more accurate superpixels than the method with $D_{\mathrm{TSPol}}^{J\mathrm{mat}} + D_{\mathrm{SLIC}}$, as shown in Figs. 14 and 16. Our method can correctly discriminate pixels from two regions with

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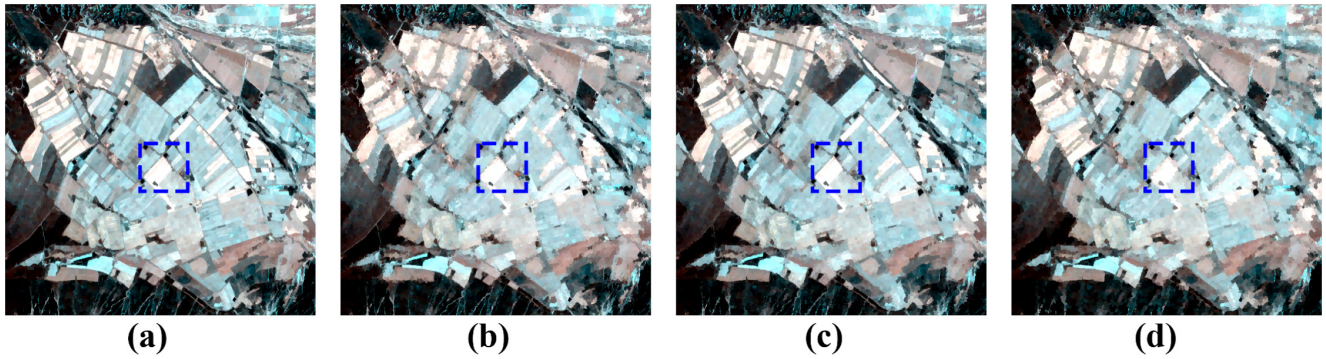

Fig. 10. Comparison of superpixel generation results ($K = 8000$) with the proposed method and the Pol-ASLIC method with the dual-polarization data. (a) Proposed method with the time-series images. (b) Pol-ASLIC method with the 20190322 image. (c) Pol-ASLIC method with the 20190813 image. (d) Pol-ASLIC method with the 20190918 image. The local results marked in blue rectangle boxes in (a)–(d) will be clearly presented in Fig. 12. Particularly, the superpixel generation results are presented with the time-series average PolSAR data.

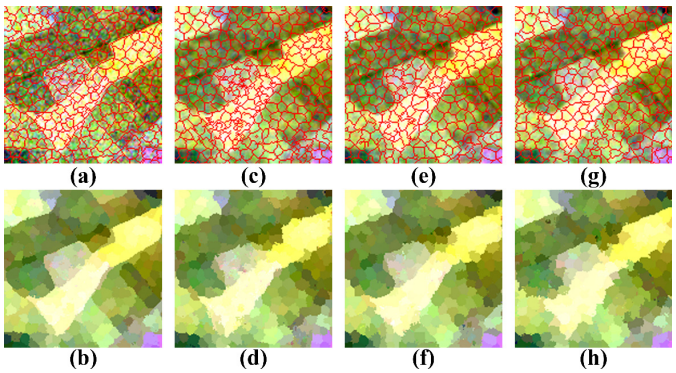

Fig. 11. Comparison of local superpixel generation results ($K = 8000$, marked in cyan rectangle boxes in Fig. 9) with the proposed method and the Pol-ASLIC method with the quad-polarization data. (a) and (b) Proposed method. (c) and (d) Pol-ASLIC method with the 20090427 image. (e) and (f) Pol-ASLIC method with the 20090521 image. (g) and (h) Pol-ASLIC method with the 20090918 image. Particularly, the generated superpixel boundaries are overplotted on the polarimetric color-coded image based on the time-series average PolSAR data.

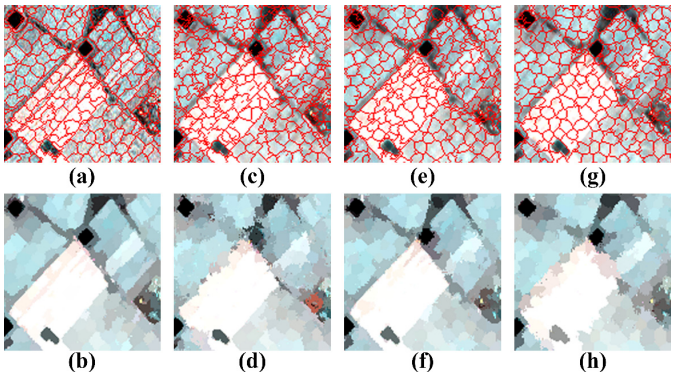

Fig. 12. Comparison of local superpixel generation results ($K = 8000$, marked in blue rectangle boxes in Fig. 10) with the proposed method and the Pol-ASLIC method with the dual-polarization data. (a) and (b) Proposed method with time-series images. (c) and (d) Pol-ASLIC method with the 20190322 image. (e) and (f) Pol-ASLIC method with the 20190813 image. (g) and (h) Pol-ASLIC method with the 20190918 image. Particularly, the generated superpixel boundaries are overplotted on the polarimetric color-coded image based on the time-series average PolSAR data.

different characteristics. Furthermore, we select two representative pixels A and B to prove the difference of such two regions, as shown in Figs. 15 and 17. Based on the average PolRGB image and the optical RGB image, such two pixels have different colors and belong to two different types. For both two study areas, pixel A belongs to the bare land, which is located outside the crop parcel, while pixel B belongs to the crop, which is located inside the parcel. Besides, their temporal evolutions of backscattering coefficients are different. So that the pixel A and B should be aligned to two different superpixels, respectively. While the method with $D_{\mathrm{TSPol}}^{J\,\mathrm{mat}} + D_{\mathrm{SLIC}}$ clusters these two pixels into one superpixel, that proves the limitations of the traditional similarity $D_{\mathrm{TSPol}}^{J\,\mathrm{mat}}$ in characterizing temporal evolutions. Although the improvement of BR of our methods compared to the method with $D_{\mathrm{TSPol}}^{J\,\mathrm{mat}} + D_{\mathrm{SLIC}}$ is only about 1% for both quad- and dual-polarization data, the sensitivity of the proposed polarimetric similarity for time-varying characteristics is helpful to promote accurate time-series PolSAR homogeneity estimation.

Second, our method is compared with the method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}^{2}$, to prove the effectiveness of the improved SLIC clustering similarity D_{SLIC}. As shown in Figs. 18 and 19, our method achieves better results in maintaining the boundaries of local superpixels than the method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}^{2}$, in the comparison of local segmentation results. Furthermore, for the quad- and dual-polarization data, the BR of our method are 1% and 4% higher than that of the method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}^{2}$. Under the same condition of time-series polarimetric similarity, these comparisons prove that the proposed D_{SLIC} is more effective than the D_{SLIC}^{2}. Their difference mainly lies in the combination ways of various kinds of similarity, where the D_{SLIC} uses the linear sum and the D_{SLIC}^{2} uses the square sum. It can be found that the way of linear sum in the D_{SLIC} may achieve better balanced effect for multiple kinds of similarities and obtains higher accuracies in our experiments, with respect to the D_{SLIC}^{2}.

Third, our method is compared with Bao's method, to present the advantage of our method compared with the traditional time-series PolSAR superpixel generation method [38]. Compared with Bao's method, there are two main improvements for our method, including the time-series polarimetric similarity and the SLIC clustering similarity.

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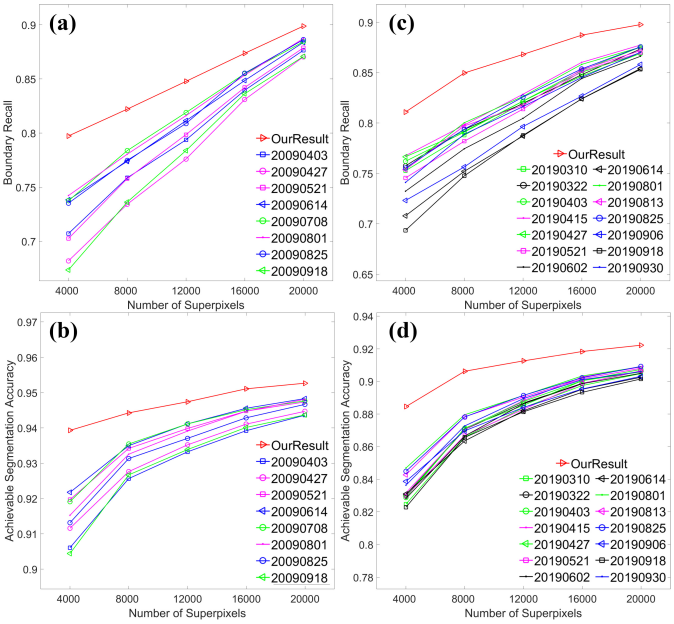

Fig. 13. Comparison of the BR and the ASA with the proposed method with time-series images and the Pol-ASLIC method using single-date image, under different number of superpixels. (a) and (b) Comparison with Radarsat-2 quad-polarization data. (c) and (d) Comparison with Sentinel-1 dual-polarization data.

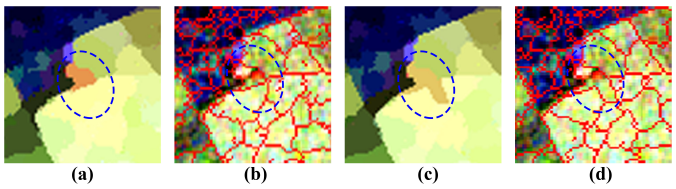

Fig. 14. Comparison of the local superpixel generation results ($K = 4000$) with the time-series quad-polarization data, to evaluate the effectiveness of the proposed time-series polarimetric similarity $D_{\mathrm{TSPol}}^{\mathrm{rms}}$. (a) and (b) Proposed method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}$. (c) and (d) Method with $D_{\mathrm{TSPol}}^{J\mathrm{mat}} + D_{\mathrm{SLIC}}$.

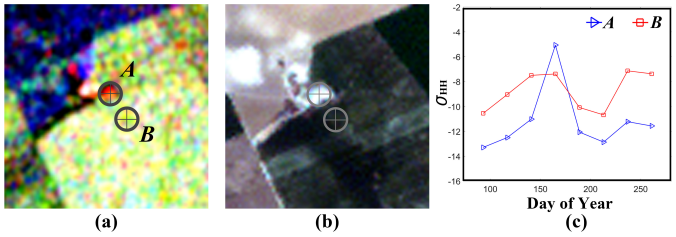

Fig. 15. Comparison of two representative pixels A and B for the time-series quad-polarization data. (a) Time-series average PolRGB image. (b) Optical RGB image derived from the RapidEye image in 20090814. (c) Comparison of time-varying feature curves.

Herein, this comparison aims to prove the effectiveness of our method compared with existing methods. The specific improvements can be referred to other discussions, for example, the improvement of time-series polarimetric similarity is shown in the comparison between our method and the method with $D_{\mathrm{TSPol}}^{J\mathrm{mat}} + D_{\mathrm{SLIC}}$, the improvement of the homogeneity factor and the time-series power similarity for the SLIC clustering is shown in the following section. As shown in Figs. 20 and 21,

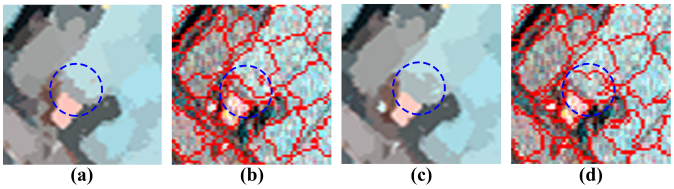

Fig. 16. Comparison of local superpixel generation results ($K = 4000$) with the time-series dual-polarization data, to evaluate the effectiveness of the proposed time-series polarimetric similarity $D_{\mathrm{TSPol}}^{\mathrm{rms}}$. (a) and (b) Proposed method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}$. (c) and (d) Method with $D_{\mathrm{TSPol}}^{J\mathrm{mat}} + D_{\mathrm{SLIC}}$.

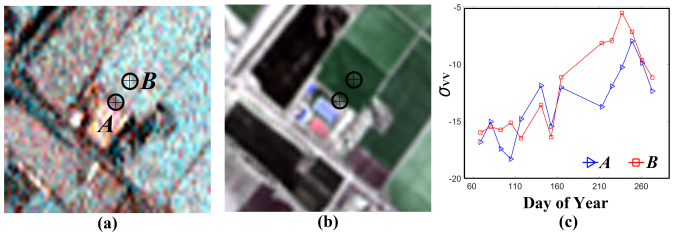

Fig. 17. Comparison of two representative pixels A and B for the time-series dual-polarization data. (a) Time-series average PolRGB image. (b) Optical RGB image derived from the Sentinel-2 image in 20190820. (c) Comparison of time-varying feature curves.

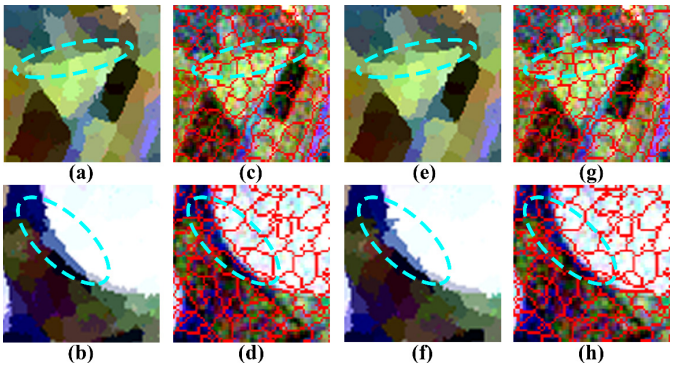

Fig. 18. Comparison of local superpixel generation results ($K = 8000$) with the time-series quad-polarization data, to evaluate the effectiveness of the proposed SLIC clustering similarity D_{SLIC}. (a)–(d) Proposed method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}$. (e)–(h) Method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}^{2}$.

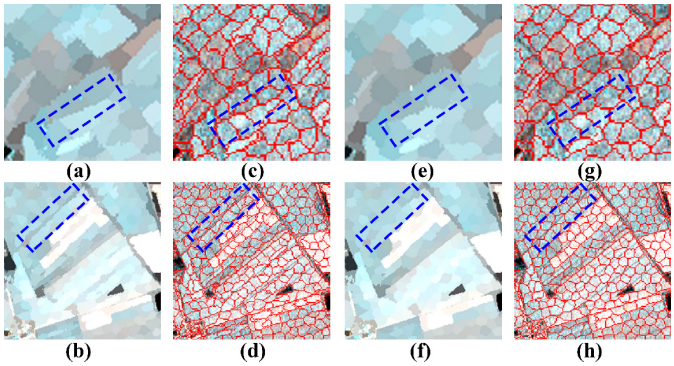

Fig. 19. Comparison of local superpixel generation results ($K = 8000$) with the time-series dual-polarization data, to evaluate the effectiveness of the proposed SLIC clustering similarity D_{SLIC}. (a)–(d) Proposed method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}$. (e)–(h) Method with $D_{\mathrm{TSPol}}^{\mathrm{rms}} + D_{\mathrm{SLIC}}^{2}$.

our method is superior to Bao's method, where our generated superpixels can adhere to the true parcel boundaries. For the quad- and dual-polarization data, the maximum improvement

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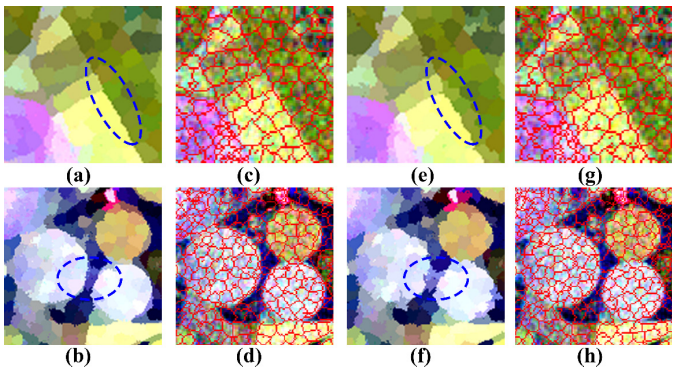

Fig. 20. Comparison of superpixel generation results ($K = 8000$) of the proposed method and Bao's method with the time-series quad-polarization data. (a)–(d) Proposed method. (e)–(h) Bao's method.

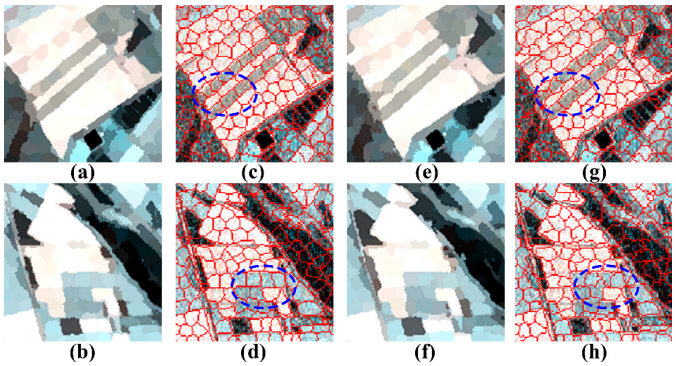

Fig. 21. Comparison of superpixel generation results ($K = 8000$) of the proposed method and Bao's method with the time-series dual-polarization data. (a)–(d) Proposed method. (e)–(h) Bao's method.

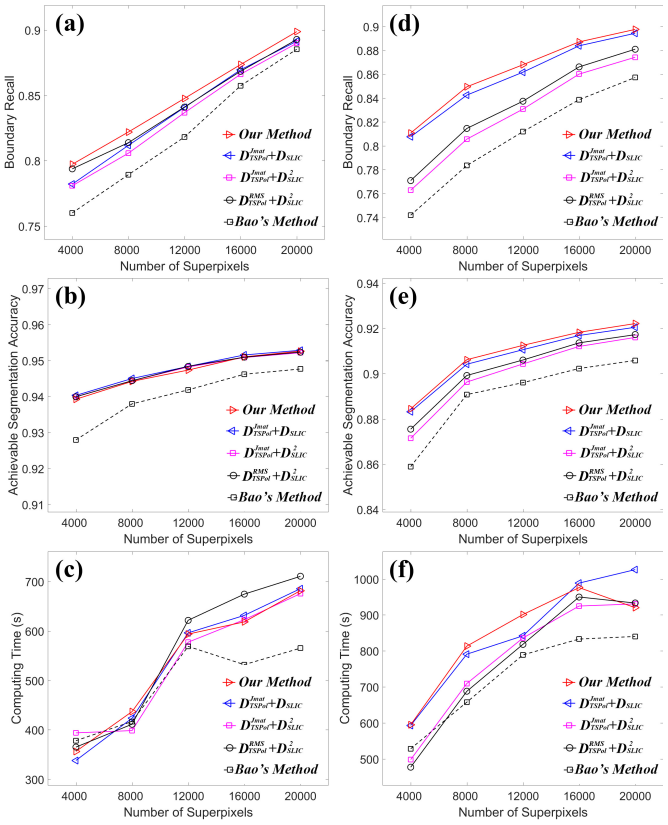

Fig. 22. Presentation of the BR, the ASA, and the computing time of five different methods. (a)–(c) Mean the comparison with the quad-polarization data. (d)–(f) Mean the comparison with the dual-polarization data.

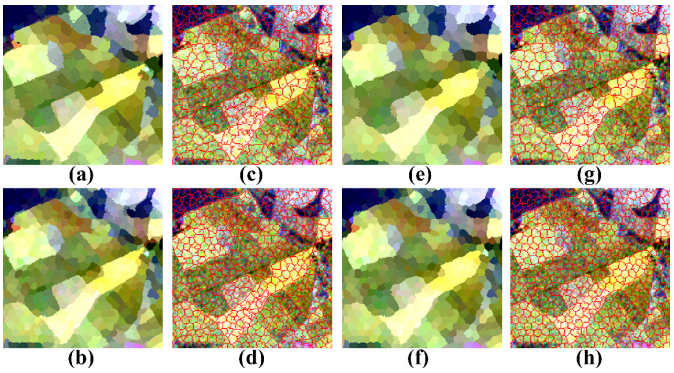

Fig. 23. Comparison of superpixel generation results with the time-series quad-polarization data. (a) and (c) Proposed method with the homogeneity factor ($K = 4000$). (b) and (d) Proposed method with the homogeneity factor ($K = 8000$). (e) and (g) Proposed method with the fixed weighting factor ($\beta_{\text{adp}} = 2$ and $K = 4000$). (f) and (h) Proposed method with the fixed weighting factor ($\beta_{\text{adp}} = 2$ and $K = 8000$).

of BR can reach 4% and 7%, respectively, and that of ASA can reach 1% and 3%, respectively.

In a word, our method performs better than other methods under different number of superpixels. As shown in Fig. 22, the BR of our generated superpixel results can reach 90% under 20 000 superpixels for both quad- and dual-polarization data. The largest computing time of our method is smaller than 700 s for eight quad-polarization images and 1000 s for 14 dual-polarization images, where the relevant programs are written in MATLAB language using a desktop with eight cores of CPU i7-4790K at 4.0 GHz. It can be demonstrated that the proposed similarity measure and superpixel segmentation method are reasonable and effective.

D. Analysis of Effectiveness of the Homogeneity Factor and the Time-Series Power Similarity

In this section, the effectiveness of the homogeneity factor and the time-series power similarity is analyzed. The tradeoff factor of the compared method is set to 2 for the verification of the self-adaptivity. As shown in Figs. 23 and 24, compared with the fixed tradeoff factor, the adaptive tradeoff factor with the homogeneity factor is beneficial for generating superpixels accurately attaching to the true boundaries. There are many over- or under-segmentation errors for the fixed tradeoff factor. Fig. 25 presents the quantitative improvements of the homogeneity factor and the time-series power similarity for the superpixel generation. It can be seen that both indicators bring about positive improvements for superpixel generation, where the improvements of the homogeneity factor are higher than that of the time-series power similarity. Furthermore, such improvements for the dual-polarization data are higher than that for the quad-polarization data. With the increasement of number of superpixels, the size of superpixel becomes smaller and it is easier to fit the boundaries of the parcels, the

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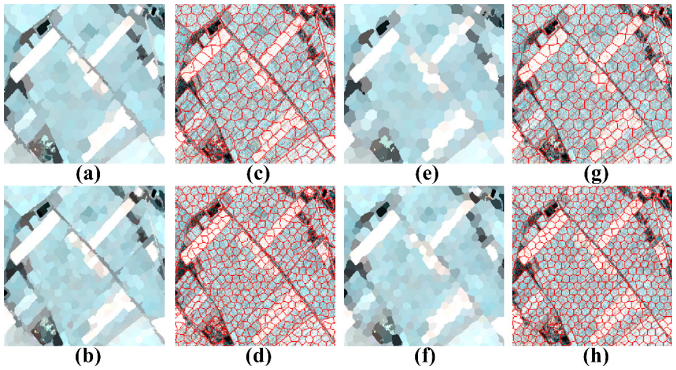

Fig. 24. Comparison of superpixel generation results with the time-series dual-polarization data. (a) and (c) Proposed method with the homogeneity factor ($K = 4000$). (b) and (d) Proposed method with the homogeneity factor ($K = 8000$). (e) and (g) Proposed method with the fixed weighting factor ($\beta_{\text{adp}} = 2$ and $K = 4000$). (f) and (h) Proposed method with the fixed weighting factor ($\beta_{\text{adp}} = 2$ and $K = 8000$).

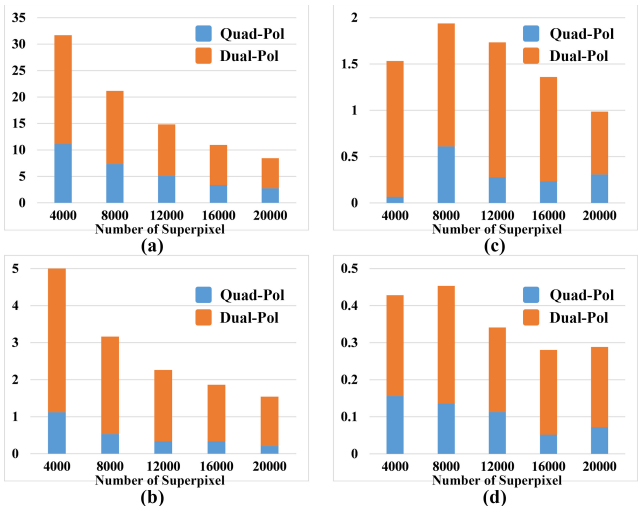

Fig. 25. Comparison of the accuracy improvement caused by the homogeneity factor and the time-series power similarity. (a) Improvement of BR by the homogeneity factor. (b) Improvement of ASA by the homogeneity factor. (c) Improvement of BR by the time-series power similarity. (d) Improvement of ASA by the time-series power similarity.

improvements of the homogeneity factor decrease. Although the improvement of the time-series power similarity is smaller than 2%, it is still helpful for guiding the SLIC clustering.

V. Conclusion

Time-series PolSAR data has been successfully applied to many fields such as time-series monitoring, change detection, and classification. However, there are two main problems for the existing time-series PolSAR superpixel generation methods. First, the traditional time-series polarimetric similarity measure based on the joint covariance matrix has limitations in discriminating different multitemporal similarity sequences with different fluctuations. It constrains the applicability of the time-series polarimetric similarity in separating targets with different temporal evolutions. Second, the existing time-series PolSAR superpixel generation methods are difficult to determine the tradeoff factor controlling the polarimetric similarity and the spatial proximity. Such two problems limit the development of the superpixel generation method, and make the time-series PolSAR data difficult to apply in the object-level mapping applications. In this article, we propose an adaptive superpixel generation method for the time-series quad- and dual-polarization SAR data, named TSPol-ASLIC. Based on the statistical characteristics of time-series PolSAR data, the proposed method has three improvements as 1) a novel time-series polarimetric similarity measure based on the rms is proposed. It enlarges the weight of the large similarity in the multitemporal similarity sequence and is beneficial to separate the sequences with different fluctuations; 2) an edge detection method based on the s2-D GS window is introduced; and 3) an improved SLIC clustering similarity is utilized. It is the linear sum of the time-series polarimetric, time-series power, spatial similarities, and the homogeneity factor. The aforementioned improvements are applied to the SLIC segmentation scheme. In order to evaluate the effectiveness of our method, eight quad-polarization SAR images of Radarsat-2 and 14 dual-polarization SAR images of Sentinel-1 are used for superpixel generation. The visual presentation and quantitative evaluation of the results prove that the edge detection and superpixel generation results of our method are better than the single-date Pol-ASLIC method. Furthermore, the utilized similarity measure and superpixel generation method perform better than those of the traditional method. Conclusively, the TSPol-ASLIC method is an effective superpixel generation method with time-series PolSAR data.

Appendix A

Related to the calculation of the conventional time-series polarimetric similarity based on the joint covariance matrix [Section II-A, (8)], we prove that

$$D_Z = \frac{1}{2}\text{Tr}\left(\hat{Z}_j^{-1}\hat{Z}_i + \hat{Z}_i^{-1}\hat{Z}_j\right) - mN = \sum_{t=1}^{N} D_{\text{SRW}}^t. \quad \text{(A1)}$$

We start by defining two block-diagonal matrices A and B as follows:

$$A = \hat{Z}_j^{-1}\hat{Z}_i \quad \text{(A2)}$$

$$B = \hat{Z}_i^{-1}\hat{Z}_j. \quad \text{(A3)}$$

Then the time-series polarimetric distance D_Z can be described as

$$D_Z = \frac{1}{2}\text{Tr}(A+B) - mN = \frac{1}{2}[\text{Tr}(A) + \text{Tr}(B)] - mN. \quad \text{(A4)}$$

Because the product operation between two block-diagonal matrices can be converted into the combination of multiple product operations of subblock matrices. The inversion operation of block-diagonal matrix is also calculated by merging multiple inversion operations of subblock matrices. The matrix A can be written as

$$A = \begin{bmatrix} \boldsymbol{a_1} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \boldsymbol{a_N} \end{bmatrix} = \begin{bmatrix} \Sigma_{1j}^{-1}\hat{\Sigma}_{1i} & \cdots & \mathbf{0} \\ \vdots & \ddots & \vdots \\ \mathbf{0} & \cdots & \Sigma_{Nj}^{-1}\hat{\Sigma}_{Ni} \end{bmatrix} \quad \text{(A5)}$$

where $\boldsymbol{a_t}$ denotes the tth subblock matrix of A. Same with such transformation of the inversion operation, the trace operation

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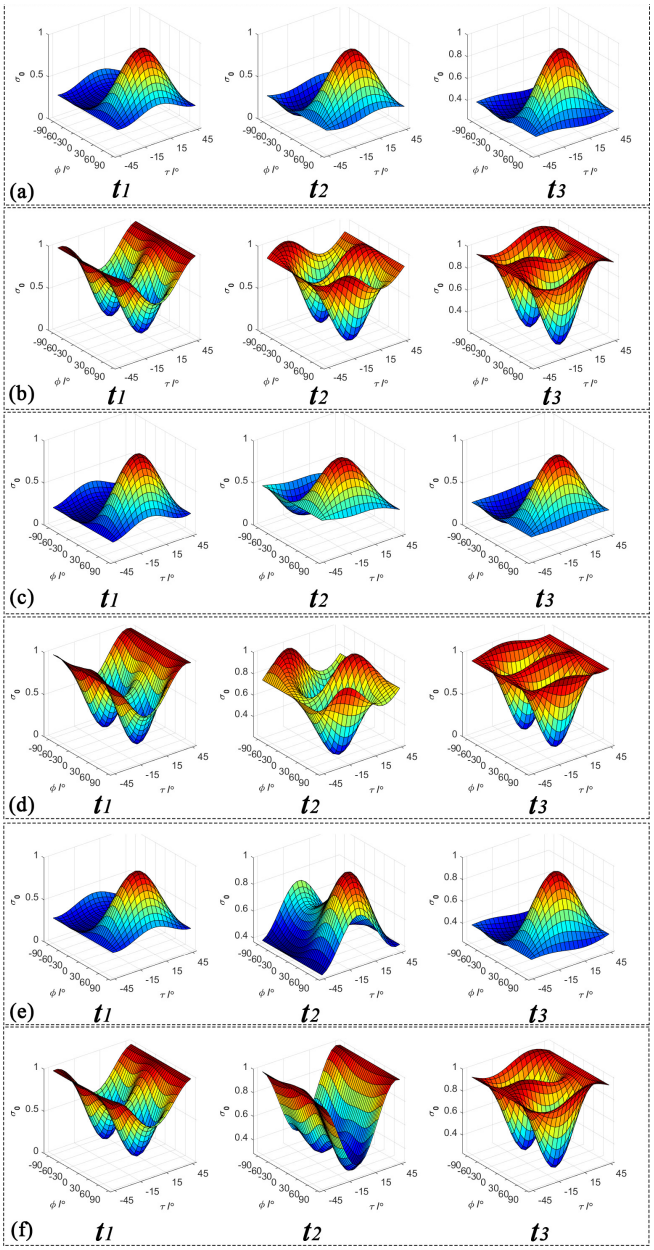

Fig. 26. Polarization signatures of P_1, P_2, and P_3. Time-series co-polarization signatures presentation of (a) P_1, (b) P_1, (c) P_2, (d) P_2, (e) P_3, and (f) P_3.

of a block-diagonal matrix can be defined by the sum of trace operations of subblock matrices. So, that the trace of A can be calculated by

$$\mathrm{Tr}(A) = \sum_{t=1}^{N} \mathrm{Tr}(\boldsymbol{a}_t) = \sum_{t=1}^{N} \mathrm{Tr}\left(\Sigma_{tj}^{-1}\hat{\Sigma}_{ti}\right). \tag{A6}$$

Meanwhile, the trace operation of B can be transformed by

$$\mathrm{Tr}(B) = \sum_{t=1}^{N} \mathrm{Tr}\left(\Sigma_{ti}^{-1}\hat{\Sigma}_{tj}\right). \tag{A7}$$

Therefore, D_Z can be expressed by

$$\begin{aligned} D_Z &= \frac{1}{2}[\mathrm{Tr}(A) + \mathrm{Tr}(B)] - mN \\ &= \sum_{t=1}^{N} \frac{1}{2}\left[\mathrm{Tr}\left(\Sigma_{tj}^{-1}\hat{\Sigma}_{ti}\right) + \mathrm{Tr}\left(\Sigma_{ti}^{-1}\hat{\Sigma}_{tj}\right)\right] - \sum_{t=1}^{N} m \\ &= \sum_{t=1}^{N}\left[\frac{1}{2}\mathrm{Tr}\left(\Sigma_{tj}^{-1}\hat{\Sigma}_{ti} + \Sigma_{ti}^{-1}\hat{\Sigma}_{tj}\right) - m\right] \\ &= \sum_{t=1}^{N} D_{\mathrm{SRW}}^{t}. \end{aligned} \tag{A8}$$

APPENDIX B

In order to explain the advantage of the $D_{\mathrm{TSPol}}^{\mathrm{rms}}$, three pixels ($P_1$, P_2, and P_3) with different time-series polarimetric covariance matrices are presented [Section II-A, Fig. 3]. The corresponding time-series polarimetric covariance matrices are visualized with the polarization signatures, shown in Fig. 26. The polarization signatures plot the normalized copolarization and crosspolarization power σ_0 under all the polarization spaces. Each polarization space can be denoted by two parameters, including the orientation angle (ϕ) with its range from $-90°$ to $90°$ and the ellipticity angle (τ) with its range from $-45°$ to $45°$. Combined with various ϕ and τ with the increment of $5°$, 648 normalized copolarization powers and 648 normalized crosspolarization powers are calculated and applied to plot the polarization signatures.

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